

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (AOC) Examination September 2019

OC 829: Synthetic Dyes and Pigments

Date: 16/09/2019 Day: Monday Time: 10.00 a.m. to 10.30 a.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q-I A Choose the correct answer for the following questions. 10

- 1 The $\lambda_{\max}$ of 1,5-diNH₂ AQ dyes is similar to
 - A. 1-NH₂AQ
 - B. 2-NH₂AQ
 - C. 3-NH₂AQ
 - D. 2,5-diNH₂AQ
- 2 The auxochromes in benzene azo resorcinol is
 - A. N=N group
 - B. N=O group
 - C. 2 OH group
 - D. A and B

- 3 The chromophore in methyl red dye:
- A. Azo group
 - B. Azoxy group
 - C. Nitroso group
 - D. A and B
- 4 Examples for strong chromophoric groups are
- A. $C=C$, $C=O$, $C=S$
 - B. NO_2 , $N=O$, $N=N$
 - C. NH_2 , NHR , NR_2 , OH
 - D. None of the above
- 5 Bromamine acid contains
- A. $-NH_2$, $-SO_3H$ and $-Br$ group
 - B. $-NH_2$ and $-OH$ group
 - C. $-NH_2$ and $-COOH$ group
 - D. None of the above
- 6 Methyl red used as
- A. laxative
 - B. antitumor agents
 - C. indicator in acid-base titrations
 - D. none of the above
- 7 Reactive dyes react with substrate by forming a
- A. Π -bond
 - B. σ -bond
 - C. hydrogen bond
 - D. ionic bond
- 8 Visible range corresponds to
- A. below $35.5 \text{ Kcal.mole}^{-1}$
 - B. $71.0\text{-}150 \text{ Kcal.mole}^{-1}$
 - C. $150\text{-}250 \text{ Kcal.mole}^{-1}$
 - D. $35.5\text{-}71.0 \text{ Kcal.mole}^{-1}$

9 The chromophore in benzene azo p-cresol is

- A. azoxy group
- B. nitroso group
- C. azo group
- D. none of the above

10 As the number of fused ring increases the colour deepens.

- A. More (from colourless to black)
- B. Less
- C. To yellow
- D. Not effected

Q-I B Draw the structure of the following dyes.

10

1 Iron green lake of fast green O

2 Auramine G

3 Naphthol blue black 6B

4 Fluorescein

5 Amido yellow E

6 Rhodamine

7 Croconium dyes

8 Polar yellow brown

9 Malachite green

10 Primuline base

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OC 829: Synthetic Dyes and Pigments

Date: 16/09/2019 Day: Monday Time: 10.30 a.m. to 01.00 p.m. Maximum Marks: 50

Important Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.
 - Make suitable assumptions and draw neat figures wherever required.
 - Use of non-programmable calculator is allowed.
 - Show necessary calculations.
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SECTION-I

Q-II Answer the following. 10

A Mention the factors which are important in the reactive dyes. 3

OR

A Explain: Oxidative hair dyes. 3

B Mention the characteristics of Fluorescence Brightener. 3

C Mention the terms involved in FBA and explain their role. 4

OR

C Give the properties of Tinopal CBS-X and Tinopal OB. 4

Q-III Answer the following. 10

A Mention the general approach to produce NIR dyes giving examples. 4

B Why 1,4 – Donar AQ is more bathochromic than 1,5 – Donar AQ. 3

OR

B Explain why 1–Donar AQ isomer of AQ is more bathochromic than 2–Donar AQ. 3

C Write note on Cyanine type chromophore. 3

SECTION-II

Q-IV Answer the following. 10

A Explain the effect of solvent on tautomerism. (in case of azo dyes) 3

OR

A Write note on Polyazodyes in which azo group is not conjugated to each other. 3

B Explain the tautomerism of hydroxyazo dyes of benzene series. 3

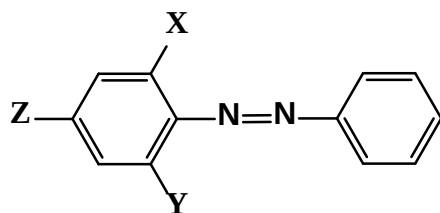
C Why tautomerism is not observed in α -phenyl azo-phenol dyes but observed in similar naphthol derivative. 4

Q-V Answer the following.**10****A** Mention the characteristics of the dye.

3

B Explain the following data for the given dyes.

3



Sr. No.	X	Y	Z	$\lambda_{\max}$ (nm)
1	H	H	NO ₂	453
2	NO ₂	H	H	425
3	NO ₂	H	NO ₂	491
4	NO ₂	NO ₂	NO ₂	520

C Write note on: chromophore theory.

4

Q-VI Answer the following.**10****A** How one can get yellow dyes in colour photography.

3

OR**A** Explain: "Brooker Deviation" giving suitable example.

3

B Explain: the dye forming process in colour photography.

3

C Write note on: The image diffusion process.

4